CHAPTER 4

AN INTELLIGENT SYSTEM TO PREDICT DIABETIC RETINAL DISEASES BASED ON DIABETIC ATTRIBUTES

4.1 OVERVIEW OF THE PROPOSED RESEARCH

Medical imaging plays a vital role in modern healthcare, offering non-invasive insights into the human body to aid in diagnosis, treatment planning, and disease monitoring. The integration of ML and DL technologies has revolutionized this domain, enabling the automation of complex tasks, enhancing diagnostic accuracy, and facilitating personalized healthcare.

ML provides models that can identify patterns and anomalies within medical images, while DL, a subsection of ML, leverages neural networks to practice huge capacities of data and achieve superior performance in image analysis. Key advancements in these fields have directed to the improvement of systems capable of segmenting, classifying, and detecting diseases with precision comparable to or even surpassing human experts.

4.2 CHALLENGES IN TRADITIONAL GLAUCOMA DETECTION METHODS

- Elevated IoP is a key risk element for glaucoma, nevertheless it does not always correlate with the disease. Some patients with normal IoP)(normotensive glaucoma) can develop optic nerve damage, making reliance on IoP alone inadequate.
- The traditional assessment of the ONH by Ophthalmologists using slit-lamp biomicroscopy is subjective. Small variations in the presence of the OD can lead to misinterpretation, especially in the early stages of glaucoma.
- Standard visual field tests, like perimetry, can be time-consuming and tedious for patients. The process requires careful attention, and the

results can be influenced by patient fatigue or misunderstanding of instructions.

- Glaucoma often progresses without symptoms until it reaches advanced stages. By the time visual field defects are detectable, significant optic nerve damage may have already occurred.
- Traditional diagnostic methods are prone to human error, both in terms of interpretation and in-patient participation. For instance, a Clinician may miss subtle variations in the ONH or fail to obtain accurate optical field data.

4.3 PROPOSED METHODOLOGY

4.3.1 Data Preprocessing

Data preprocessing is a data preparation mining method of data preparation involves organizing unprocessed data in a comprehensible way. Existent world data are typically incomplete, inconsistent, devoid of particular behaviors, and riddled with errors. Preprocessing data is a quick but efficient technique to deal with these problems. Preprocessing normally contains the subsequent processes,

- Data association involves acquiring, choosing, and assimilating data about diabetic patients.
- Data cleaning comprises replacing values and eradicating replicas from the new dataset.
- Normalize data, discretized data, and construct features in the dataset.
- Data drop means of reducing dimensions.

Before entering into ML algorithms, the obtained data are cleansed and preprocessed. The cleaned dataset mentioned above is now used for additional processing. Missing row values in this case are removed from the dataset. The blood pressure and blood sugar levels are restored using the attribute’s mean values. An identifiable individual trait or attribute is meant by features. In this context both categorical and numeric features are possible. As a set of values or a boolean set is referred as categorical data. Integer values or a value sequence are both examples of numerical features. There are both types of features in the current dataset.

4.3.2 Dataset Splitting

Current appropriate set of information are divided into a train and a test data. A system uses a set of data called training data to learn, how to use tools like neural networks to learn and generate sophisticated outputs.

Table 4.1 Listing of Few Parameters

Features	Chance	No chance
BP	> 130	<= 130
BS	>120	<= 120
Diabetic years	>10	<= 10
Patient_age	>40	<=40
DM	1	0
IoP	>21 mm Hg	10–21 mm Hg
Vision acuity	1	0
Peripheral	1	0
Center	1	0
Eye pain	1	0
Headache severity	1	0
Cataract	1	0
Red eye	1	0
Smoking	1	0
Vomiting	1	0
Cortcosteriods intake	1	0

Table 4.1 describes the list of parameters and features which defines the chance of ideal and actual values. This is usually the result of a very simple model. 80% of the data is considered for the best prediction in the learner dataset. To create a decision support model, the remaining 10% of the facts is used for validation and 10% for checks.

4.3.3 ML Classification Approach

Statistics defines classification as determining which categories a given entity falls based on observations. ML classifiers are incredibly helpful for automating operations to save time and money. A classifier is a machine-based algorithm that uses certain principles to classify data. A categorization model is the product of classifier's ML. Data are sorted after the model has been trained using the classifier. Both known and unlabeled classifiers are accessible. The unlabeled datasets that unsupervised ML classifiers are fed must be categorized using anomalies, data structures, and pattern recognition. Supervised and semi-supervised classifiers are taught how to classify the data into predetermined groups using training datasets. There are only a few technical terms that are discussed. In order to resolve a predictive modeling issue, a learning instance a data fact from a knowledge set is used. Sometimes, the phrases training occurrence and training sample are used synonymously. In the context of ML, the words hypothesis and model are frequently used interchangeably. Passing on categorical distinct value class labels to data points is the foundation of a learner predictor.

On the other hand, a classifier and a hypothesis are not exactly the same thing. The method of learning is a collection of guidelines designed to identify or approximate the target function by modeling the target utility using our drill dataset. The set of thinkable hypothesis that a learning process can generate in order to build the selected final hypothesis and represent an unknown objective is known as the hypothesis space. The terms listed below are widely employed when using ML techniques to find an effective machine fit model for retinal sickness prediction.

4.3.3.1 Ensemble Based Algorithms

An Ensemble RFML Algorithm offers more improvement over bagged trees than little non corresponding trees. A random forecast trial is chosen after the whole set of predictors at the moment of tree splitting. This indicates that an average of less associated trees is used to make predictions. By choosing random predictors from each division, a random tree’s hyperparameter tuning is assembled. Less bias and less variation are produced as a result of this learning. There are two groups that are boosting and bagging. Bagging is the technique of training a single model parallel on a small subset at random. Boosting is the reverse of bagging, which uses an individual model with sequential execution and learns from errors in the past. By following the procedures, Ensemble RF created a decision tree independently while utilizing bagging as a strategy, yielding a result of 0.973 accuracy of below steps,

- Sufficient random selection.
- Construction of each decision tree sample.
- Forecast based on the highest percentage of sample votes

i) Adaboost Random Forest Ensemble Learner

This is a component of boosting ensemble, which builds a decision tree using weights from misclassified data in order to learn from past errors. The steps to implement this strategy are stated below.

- Calculate the ensemble choice tree weight.
- The first DT weighted error rate (decision tree)
- . Decision tree training.

$$Treeweight = treelearninggrate * log\left(\frac{1-e}{e}\right) \tag{4.1}$$

- Update weights to the misclassified points.
- Repeatedly do the steps.
- Prediction of model.

From the listed steps, final pronouncement is concurrence to influence of elevated tree weight.

ii) Gradient Boost Random Forest Ensemble Learner

This is a boosting method as well, however it directly uses residual error in place of weighted points. By adding every projected tree, it is creating a new prediction. To obtain the prediction, follow the steps given below.

- Train a decision tree.
- Prediction of decision tree.
- Finding the residual fault of the tree and append new.
- Until repeat the stages to obtain the determined number of tree.
- Final model.

iii) Tri Layer Neural Network

This trilayer neural network involves applying a sequence of matrix operations layer by layer and the input data is processed. Then, bias added to the product of the input and the weights for each of three layers. Then include chosen activation function through this output. The subsequent layer then uses the result of this activation function as an input and follows the same process. Since there are three levels, this operation is iterated three times. Our final step predicts which patient have chance of retinal disease from health information risk factors. The forward propagation process has come to an end. The difference between proposed prediction and the anticipated output is then calculated. This error value is used during backpropagation.

After mathematically push system error, which is the discrepancy between prediction and expected, back through the Neural Network (NN) in the opposite way during backpropagation as a result of our errors. Next, to determine what value should assign for the weights in imperative to get the finest prediction by taking the derivative of the functions. Then essentially want to understand the relationship between the weight's value and the error derived from it. The output of the node is derived by activation transfer functions.

Since the sigmoid function exists between 0 and 1, this serves as the main justification for its use. Consequently, modeling whose output is a probability prediction finds special application in this way. Since everything has a probability that only ranges from 0 to 1, the sigmoid is the optimal choice. It could serve a variety of purposes. It can therefore ascertain the slope of the sigmoid curve between any two sites. Hyperbolic tangent is comparable to an improved logistic sigmoid. From (-1) to (+1) is the range of the *tan h* function. Tan *h* also has a sigmoidal (*s*-shaped) form. Even so, the function is monotonous. Because of the logistic sigmoid function, neurons utilized for feed forward networks may become stuck during training. As of right now, the best often activation function used in worldwide exists rectified activation function. Since almost all DL algorithms and convolutional neural networks use it. The ReLU is only partially fixed, as may be observed (from underneath). F(z)=zero when z is less than zero and equals z when z is superior than or equal to zero. Both the ReLU and the derivative are monotonic. To determine the best prediction rate, all three of the suggested activations were put into practice and their outcomes are compared.

4.4 RESULTS AND DISCUSSIONS

The output measures of diverse trilayered neural network with various transfer functions. In order to escalating the recital standards, this recommended system is hosted the NN in the type of trilayer and ReLU. When compared transfer activation functions trilayer with ReLU is giving better performance accuracy. Narrow NN is achieving the accuracy of 0.925 and append with ReLU, it brings into the improved accuracy of 0.025, means 0.95.

Figure 4.1 measures for the use of different machine learners' methods. Due to its activation functions in the data , bilayered NN has the lowest performance value in this case. Therefore, this model's is not good fit. Again, this dataset is included in ensemble, bilayered, and trilayered NN, though its correctness are only reasonable and drop in real-time justification. Nevertheless, Trilayer NN with ReLU is capable to detect and project diabetic retinal chaos in diabetic folks with a fine level of accurateness.

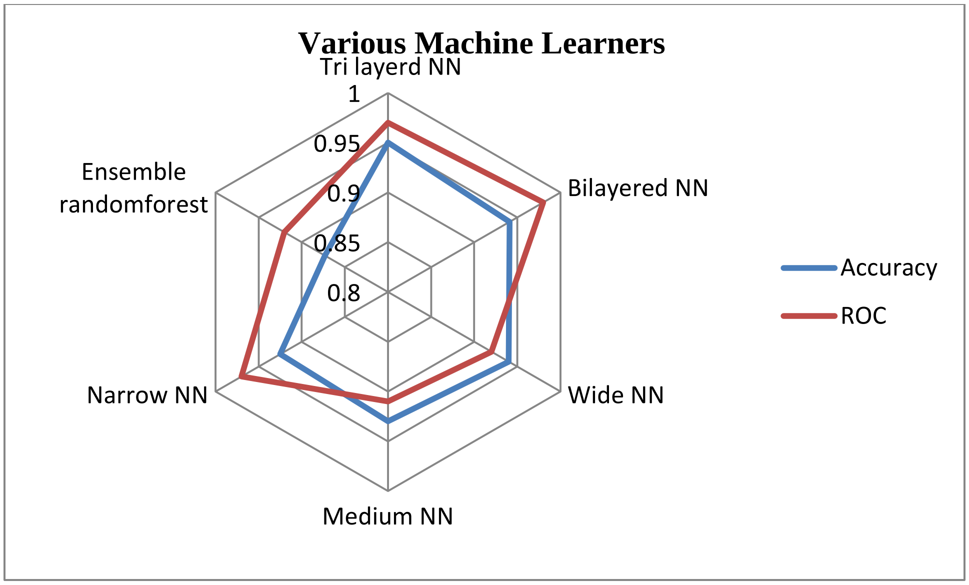

Figure 4.1 Comparisons of Various Machine Learners

With a grouping rate on possibility of 0.0216 and non-possibility value 0.210, an indecision matrix of the learning data reviewed with a self-belief level of 0.91 reached a rate of misclassification of 0.21. The paramount fit for forecast diabetic retinal disease based on presentation capability is an assertive.

Table 4.2 Diabetic Retinal Disease Predictions on Diabetic Mellitus Persons

DPid	**Forecast**		**Expert 1**	**Expert 2**
	Occur	**Not occur**		
02	0.967	0.023	Correct	Moderate
13	0.874	0.126	Correct	Correct
14	0.105	0.985	Moderate	Moderate

Table 4.2 represents the Dpid14 has been very modest likelihood of vision disease, since the individual having normal blood value of sugar, standard level of pressure in blood , IoP abnormal, cdr normal and absence of DM . But the patientid 02 had more risk of retinal disease. The patient seeks an immediate responsiveness to meet up the experts or ophthalmologists, due to high risk

association factor predicted to patientid 13. This patient has been with prolong 20 years of DM, all along with high blood sugar, hyper tension at the age of 55, so that enduring also necessary professional's implication to evade visionless in prospect.

4.5 SUMMARY

A novel clinical dataset has been developed for reliable data analysis. This dataset incorporates carefully preprocessed data with relevant features for advanced analysis. This approach utilizes a diverse set of feature combinations to effectively predict diabetes-related vision impairments, including DR, glaucoma, and DME. By employing multiple classifiers, specialists can rapidly identify these conditions. This system will be disseminated to Ophthalmologists, frontline healthcare providers, diabetic patients, besides other relevant stakeholders to facilitate early detection of diabetic retinal diseases during routine examinations.